INTERNSHIP REPORT

A report submitted in partial fulfillment of the requirements for the Award of Degree of

BACHELOR OF ENGINEERING
in
COMPUTER ENGINEERING
by Student

Tarun Sanjay Pardeshi
(72313104L)

Under Supervision
of

Cognifyz Technologies Internship
(Duration: 27 January 2026 – 27 February 2026)

DEPARTMENT OF COMPUTER ENGINEERING

Modern Education Society's College of Engineering ,Pune Approved by AICTE,

Affiliated to SPPU, Pune

Maharashtra
DEPARTMENT OF COMPUTER
ENGINEERING MODERN EDUCATION
SOCIETY'S COLLEGE OF ENGINEERING ,PUNE

CERTIFICATE

This is to certify that the "**Internship report"** submitted by **Tarun Sanjay Pardeshi (72313104L)** and **PRN** is **F23113015** is work done by him and submitted during 2025-26 academic year, in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF ENGINEERING in COMPUTER ENGINEERING,** at **Cognifyz Technologies Internship.**

College Internship Coordinator **Department Internship Coordinator**

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Head of the
Department of
Computer Engineering

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Cognifyz Technologies

Internship Completion Certificate

Date -26/02/2026

This is to certify that **Tarun Sanjay Pardeshi, (Intern ID: CTI/A1/C298496)**, currently pursuing a B.E. from The Modern Education Society Wadia College of Engineering, was working as a **Data Analysis Intern** with Cognifyz Technologies from 27/01/2026 to 27/02/2026.

During this period, he has served as a Data Analysis Intern and has displayed remarkable dedication, sincerity, and a strong desire to learn. He has exhibited exceptional coordination skills and effective communication abilities. Moreover, his attention to detail has been truly impressive.

He has consistently approached new assignments and challenges with enthusiasm, showcasing his passion for Data Analysis. His commitment and willingness to acquire new knowledge and skills have been evident throughout his internship.

We extend our best wishes to Tarun Sanjay Pardeshi for a successful future, and we have no doubt that he will continue to excel in the field of Data Analysis.

With Regards,
Cognifyz Technologies

Cognifyz Technologies
Cognifyz Technologies
Nagpur, India

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ACKNOWLEDGEMENT

I would like to express my Sincere gratitude to **Cognifyz Technologies** for providing the opportunity to work as a Data Analysis Intern and to gain practical exposure in data cleaning, exploratory analysis, and insight generation from real datasets. Appreciation is also extended to **Modern Education Society's Wadia College of Engineering, Pune** for academic support and encouragement to participate in such industry-oriented learning experiences.

This internship helped strengthen practical understanding of Python-based data analysis, preprocessing workflows, and report preparation using a real restaurant dataset. The project also improved analytical thinking, data interpretation skills, and the ability to convert raw data into meaningful business insights.

This internship has helped me enhance my technical skills as well as my analytical thinking and problem-solving abilities. It has also improved my confidence in working on real-world datasets and presenting insights through data visualization.

I am truly grateful for this opportunity, which has contributed significantly to my academic growth and career development in the field of data analytics.

Tarun Sanjay Pardeshi
(72313104L)

ABSTRACT

This report presents the internship work carried out at Cognifyz Technologies on the project **Restaurant Data Analysis**. The project used a restaurant dataset containing **9551 records**, spanning **141 cities** and **15 country codes**, with fields related to restaurant identity, cuisines, prices, ratings, and service features.

The internship work focused on data cleaning, preprocessing, exploratory data analysis, and interpretation of patterns related to ratings, location, online delivery, and table booking. The notebook evidence shows the dataset was initially loaded with shape **(9551, 21)** and then transformed for analysis by standardizing important fields into machine-readable form.

The analysis generated useful findings on city performance, price-segment behavior, and service preferences among restaurants. These findings show how restaurant data can support business understanding and decision-making through a structured analytics workflow.

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WEEKLY OVERVIEW OF INTERNSHIP ACTIVITIES

FIRST WEEK	DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED
	27/01/2026	Tuesday	Introduction to Cognifyz Technologies internship program and understanding the restaurant dataset structure
	28/01/2026	Wednesday	Loading the CSV dataset and examining columns, records, and raw values in Jupyter Notebook
	29/01/2026	Thursday	Identifying missing values, reviewing ratings, and checking service-related columns
	30/01/2026	Friday	Understanding city, country, cuisine, and price range features for further analysis
	31/01/2026	Saturday	Preparing an initial data cleaning workflow in the notebook

SECOND WEEK	DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED
	3/02/2026	Tuesday	Data cleaning and standardization of binary columns such as table booking and online delivery
	4/02/2026	Wednesday	Converting aggregate ratings into clean analytical format for grouping and summaries
	5/02/2026	Thursday	Exploring city-wise counts and average ratings
	6/02/2026	Friday	Understanding price-range distribution and service adoption patterns
	7/02/2026	Saturday	Reviewing restaurant features for meaningful business-oriented analysis

THIRD W	DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED
	10/02/2026	Tuesday	Performing exploratory data analysis on ratings, prices, and service options
	11/02/2026	Wednesday	Comparing online delivery and table booking across price segments

EEK	12/02/2026	Thursday	Identifying highly rated cities and top-performing restaurant markets
	13/02/2026	Friday	Analyzing country-level restaurant counts and ratings
	14/02/2026	Saturday	Summarizing intermediate findings for final reporting

	DATE	**DAY**	**NAME OF THE TOPIC/MODULE COMPLETED**
FOURTH WEEK	17/02/2026	Tuesday	Consolidating insights from exploratory analysis into structured findings
	18/02/2026	Wednesday	Preparing business interpretations from city and price-range analysis
	19/02/2026	Thursday	Drafting internship report sections including introduction, tasks, and analysis
	20/02/2026	Friday	Finalizing report structure based on internship project work
	26/02/2026	Thursday	Internship completion certificate issued by Cognifyz Technologies

1. INTRODUCTION

The internship at **Cognifyz Technologies** was completed in the role of **Data Analysis Intern** from **27 January 2026 to 27 February 2026**. During this internship, the assigned project focused on restaurant data analysis using a structured CSV dataset and a Python-based workflow implemented in Jupyter Notebook.

The main purpose of the project was to analyze restaurant information related to ratings, cuisines, locations, price segments, and service features such as online delivery and table booking. The project required cleaning the data, transforming categorical values into usable analytical form, and generating summaries that could reveal practical insights for restaurant business analysis.

Objectives of the Project:

- To analyze restaurant data and understand patterns in ratings, cities, and price segments.
- To perform data cleaning and preprocessing for reliable analysis
- To study the relationship between service features and price range.
- To identify top-performing cities based on ratings.
- To present the findings in a structured internship report aligned with the Cognifyz Technologies internship context.

This internship helped me bridge the gap between theoretical knowledge and practical application in the field of data analytics. I learned how to handle real-world datasets, perform analysis efficiently, and present insights in a structured and meaningful way.

Moreover, I developed important skills such as analytical thinking, problem-solving, and data interpretation.

This internship helped connect theoretical knowledge of data analysis with practical implementation using a real dataset. It also improved technical confidence in using Python-based tools for exploratory analysis and reporting

1.1 Task 1: Data Collection and Cleaning

Problem Statement

The restaurant dataset contained raw information distributed across multiple attributes such as city, cuisine, cost, ratings, and service options. Raw datasets often contain inconsistencies, unprocessed text values, and fields that are not directly suitable for analytical grouping or comparison.

Objectives

- To load the restaurant dataset into the analysis environment
- To inspect the dimensions and structure of the dataset before analysis
- To check whether the values in service and rating fields were ready for further processing.

Motivation

Data cleaning is essential because incorrect or inconsistent input values can reduce the reliability of analytical outcomes. A clean dataset provides a stronger foundation for accurate grouping, summarization, and pattern discovery.

Key Features Implemented

- Loaded the restaurant dataset in Jupyter Notebook and confirmed the working shape as **(9551, 21)**
- Reviewed major features including city, cuisines, average cost, ratings, and votes.
- Prepared the data for transformation and exploratory analysis

1.2 Task 2: Data Preprocessing and Transformation

Problem Statement

Although the raw dataset contained useful fields, service-related columns such as table booking and online delivery were stored as text values and could not directly support numerical analysis. Aggregate rating values also required clean numeric handling for grouping and summaries.

Objectives

• To standardize categorical service values into machine-readable numerical form.
• To improve analytical consistency for rating-based operations

• To prepare the dataset for city-wise and price-wise analysis

Motivation

Transformation helps convert raw data into a structured form that supports faster and more meaningful analysis. It also reduces ambiguity in service-related comparisons and improves reproducibility of results

Key Features Implemented

• Converted Yes/No fields into binary analytical form for selected service columns.
• Prepared cleaned rating values for summaries and comparisons.
• Enabled grouped analysis by price range, city, and country code.

1.3 Task 3: Exploratory Data Analysis

Problem Statement

After preprocessing, the dataset needed to be explored in order to identify performance patterns and business-relevant relationships. Raw values alone could not communicate which cities performed best or how service features differed across price ranges

Objectives

- To study city-wise average ratings and restaurant counts.
- To analyze price-range behavior in online delivery and table booking.
- To summarize country-level restaurant counts and rating behavior.

Motivation

Exploratory data analysis helps reveal patterns, outliers, and associations within the dataset. These observations guide final insights and help transform data into actionable understanding.

Key Features Implemented

- Computed city-level ranking of restaurants based on average rating.
- Produced a price-range summary including restaurant counts, delivery rate, booking rate, and average rating.
- Generated country-level summary tables from the dataset

1.4 Task 4: Result Interpretation and Insight Generation

Problem Statement

Analytical output becomes useful only when it is converted into understandable findings that explain restaurant market behavior. There was a need to interpret the computed summaries in terms of pricing, service orientation, and city-level performance

Objectives

• To interpret city and price-related summaries meaningfully.
• To identify differences between budget and premium restaurant service patterns.

• To convert numerical summaries into report-ready observations.

Motivation

Insight generation is important because businesses benefit more from clear interpretations than from raw numbers alone. Proper interpretation enables better understanding of customer-facing service choices and market variation across cities.

Key Features Implemented

• Identified that high price ranges show stronger table-booking behavior than lower price ranges
• Noted that lower price ranges show stronger online-delivery orientation than premium segments
• Highlighted the presence of highly rated city clusters in the data.

1.5 Task 5: Report Preparation and Documentation

Problem Statement

After analysis, the project findings needed to be documented in a formal internship-report structure suitable for academic submission. A structured report was necessary to present objectives, methods, analysis, results, and conclusions in a professional format.

Objectives

• To organize the project into a formal internship-report structure
• To align the report with the internship details from Cognifyz Technologies.
• To present analysis results in a clear and readable format

Motivation

Documentation is a vital part of data-analysis work because it communicates project value, methodology, and outcomes clearly to academic or professional reviewers. A proper report also demonstrates understanding beyond code execution.

Key Features Implemented

• Structured the project as a complete internship report.
• Incorporated organization details from the Cognifyz Technologies certificate.
• Summarized analytical results using dataset-derived outputs.

2. Analysis

Throughout the internship at Cognifyz Technologies, the restaurant data analysis project was carried out as a complete workflow covering data loading, cleaning, transformation, exploratory analysis, and report preparation. The objective of the project was not only to inspect the dataset but also to derive useful patterns from restaurant ratings, prices, and services.

The dataset contains **9551 restaurants**, **141 cities**, and **15 country codes**. This breadth makes the dataset suitable for comparing markets, price segments, and service practices across multiple locations.

The computed city summary places **Inner City** at the top among rated cities with an average rating of **4.9**. The next high-performing city in the generated top-cities table is **Quezon City** with an average rating of **4.8** across **1** rated restaurants.

Price-range analysis shows distinct differences in service model across segments. For **price range 1**, the table-booking rate is **0.02 percent** and the online-delivery rate is **15.77 percent**. For **price range 4**, the table-booking rate rises to **46.76 percent**, while the online-delivery rate is **9.04 percent**.

These values indicate that lower-priced restaurants are more delivery-focused, while premium restaurants rely more on reservation-based dining service. This business pattern is consistent with the notebook's earlier summary where higher price ranges showed stronger table-booking usage and lower price ranges showed broader delivery adoption.

Country-level analysis also shows that the largest country-code segment in the generated summary has **Country Code 1** with **8652 restaurants** and an average rating of **2.523**. This indicates that restaurant density is not evenly distributed across country codes in the available dataset.

3. Software requirements specifications

3.1 System Configuration

Software Requirements:

• Operating System: Windows 10 / Windows 11 or equivalent analysis environment.
• Tool Used: Jupyter Notebook / Python analysis workflow.

• Data Source: Restaurant CSV dataset

Technologies Used:

• Data Analysis: Python and pandas-based processing
• Data Exploration: Grouping, aggregation, and summary generation.
• Documentation: Markdown-based structured report preparation.

3.2 Hardware Requirements:

• System: Intel Core i5 Processor or higher.
• Hard Disk : Minimum 256 GB
• RAM : Minimum 8 GB

4. Methodological Details

The internship was structured as a real-world **data analytics project lifecycle**, where I progressed through multiple stages of data preparation, analysis,cleaning,and getting meaningful insights

4.1 Weekly Iteration Rhythm

The internship followed a staged learning pattern beginning with data understanding, followed by cleaning, exploratory analysis, summary generation, and final report preparation. Each stage built on the previous one and helped create a complete workflow from raw data to final conclusions.

4.2 Data Pipeline Process

Data Collection and Loading:
The CSV dataset was loaded into Jupyter Notebook for exploration and preprocessing
Data Cleaning:
Service columns and rating values were standardized for easier analysis.
Transformation
Cleaned fields were prepared for grouping and summarization by city, price

Exploratory Data Analysis: Summary tables were created to inspect ratings, delivery behavior, and location-level performance**.**

Reporting: Insights were documented in formal internship-report format.

4.3 Exploratory Data Analysis (EDA)

EDA focused on identifying restaurant counts, average ratings, and service patterns across dimensions such as city and price range. This helped reveal which locations were strongly rated and which price segments were more delivery-oriented or booking-oriented.

4.4 Validation and Review

The analysis workflow was reviewed through repeated notebook-based checks and generated summary tables for consistency. This reduced ambiguity in the final interpretation of restaurant-service behavior and rating patterns.

4.5 Reporting and Documentation

The final stage involved structuring the project according to an internship-report format similar to the reference report while keeping the content aligned with the Cognifyz Technologies internship and the restaurant-data project

5 CODING / PYTHON OPERATIONS

Since this project was based on restaurant data analysis, the implementation primarily involved preprocessing and summary generation rather than application development. The main operations included data loading, feature cleaning, service-column transformation, and grouped summaries by city, country code, and price range.

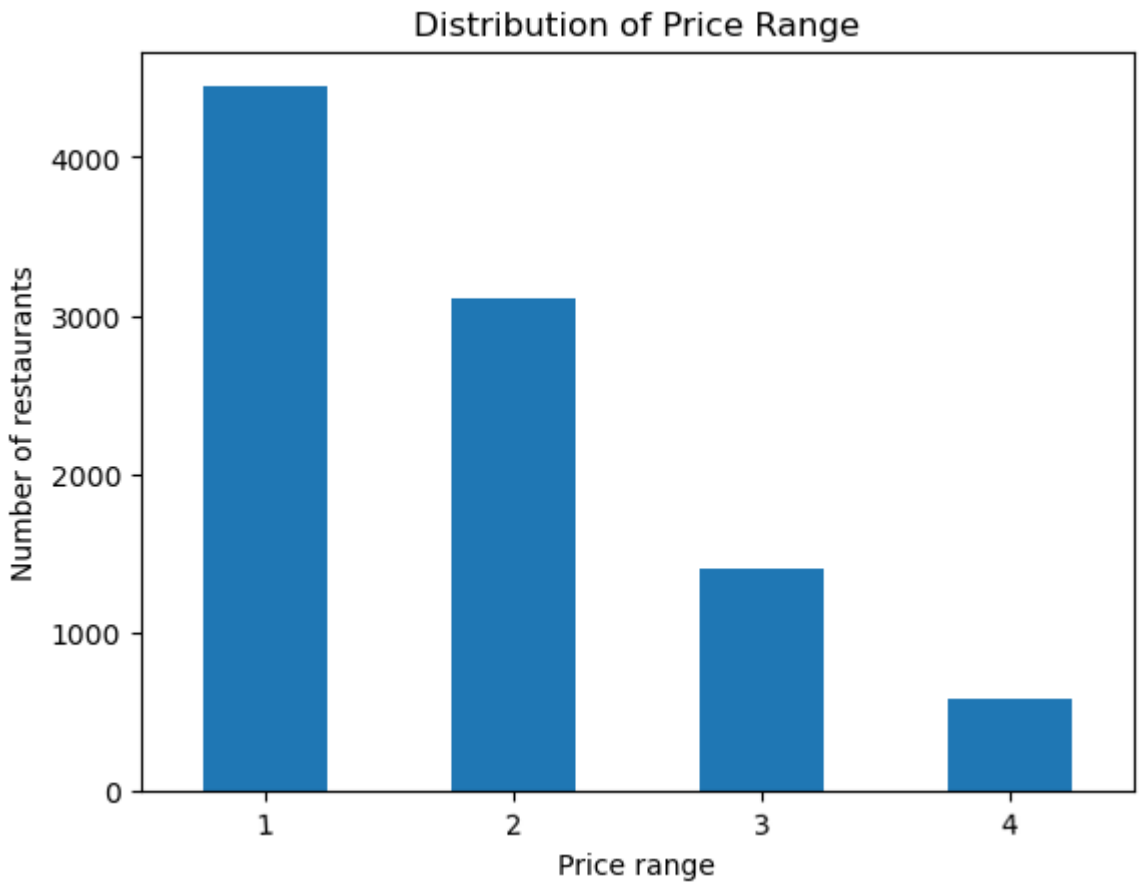
Core Functional Components

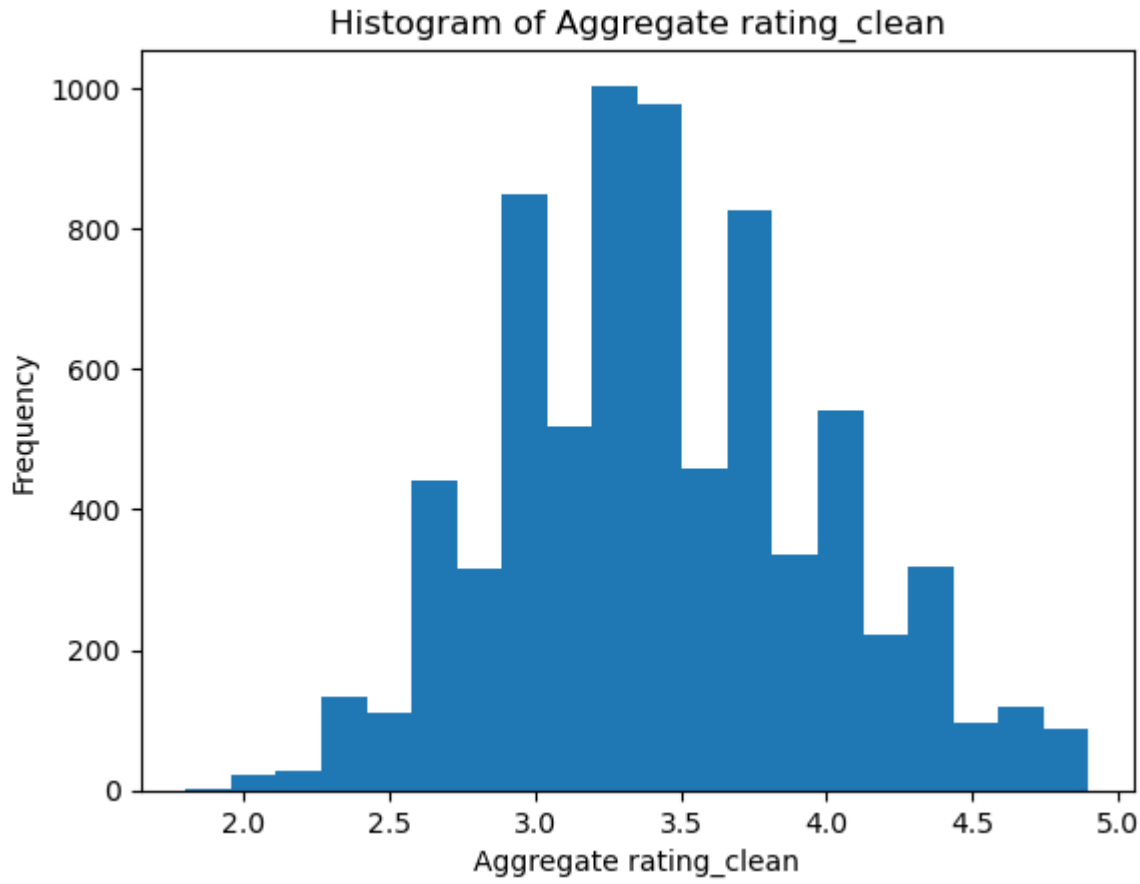
- Data Import: Loaded the restaurant CSV dataset into the notebook environment.
- Data Cleaning: Standardized service columns and handled analytical formatting for rating-related fields.
- Aggregation: Generated grouped summaries for cities, countries, and price ranges.
- Interpretation: Converted summary values into report-ready findings.

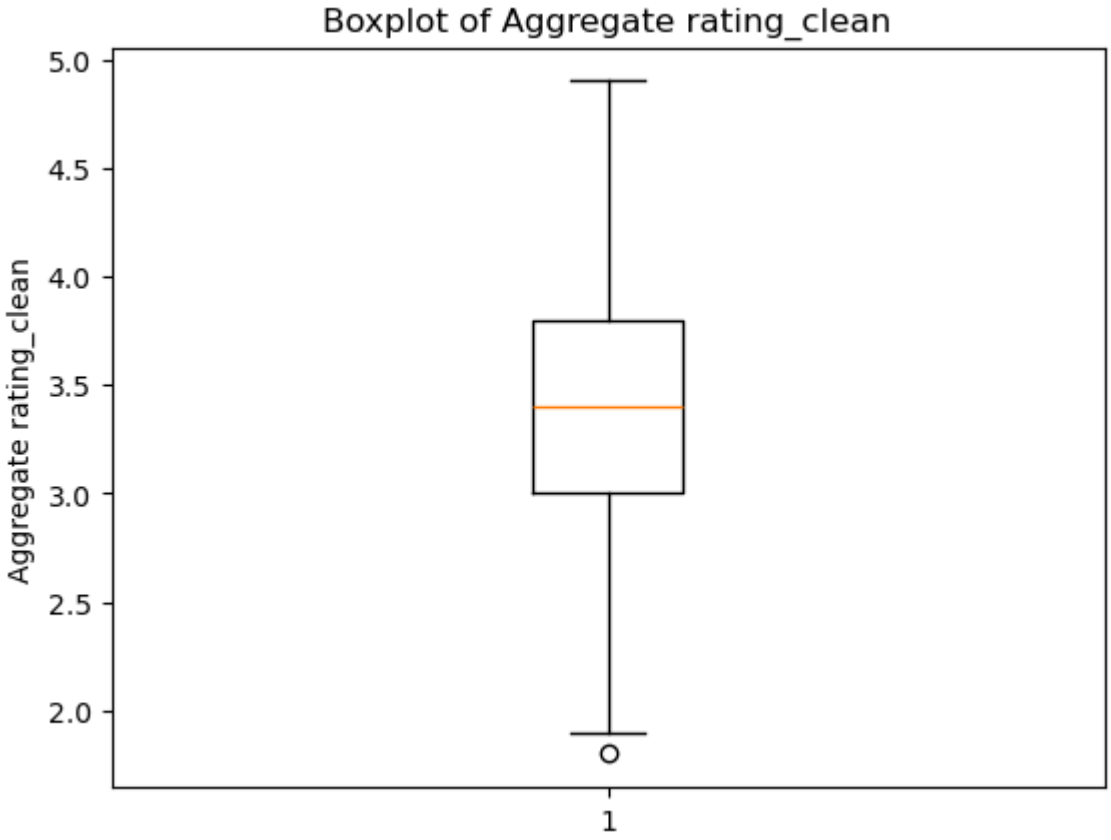
Sample Python-Based Operations

- Grouped cities by restaurant count and average rating.
- Grouped price ranges by booking rate, delivery rate, and average rating.
- Grouped country codes by total restaurants and mean rating.

6 Screenshots






Boxplot of Aggregate rating_clean
Aggregate rating_clean
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1

7. Result

The internship project produced a meaningful analysis of restaurant data by combining preprocessing and exploratory analysis into one structured workflow. The dataset-driven outputs showed that the restaurant data contains clear differences between cities and price segments.

Key Outcomes:

- A structured dataset of **9551 restaurant records** was analyzed successfully.
- The dataset covered **141 cities** and **15 country codes**, supporting broad comparison.
- The highest-rated city in the generated summary was **Inner City** with an average rating of **4.9**.
- **Price range 4** showed strong table-booking availability at **46.76 percent**.
- **Price range 1** showed stronger online-delivery orientation at **15.77 percent**.
- Country-level grouping showed concentration of restaurants within specific country-code segments.

These findings demonstrate that restaurant data can be used to understand market structure, service orientation, and rating behavior through relatively simple but effective data-analysis methods

8. Conclusion

The internship completed at Cognifyz Technologies provided valuable practical exposure to data analysis through the restaurant data analysis project. The project covered the complete flow from data loading and cleaning to exploratory analysis, interpretation, and formal report preparation.

The work showed that restaurant-service behavior differs across price ranges and that city-level rating patterns can be extracted through structured summaries. It also demonstrated the usefulness of Python and Jupyter Notebook for converting raw CSV data into meaningful business insights.

This internship strengthened technical knowledge in data preprocessing, aggregation, and reporting, while also reflecting the dedication, learning attitude, and attention to detail described in the Cognifyz Technologies internship completion certificate. Overall, the internship has contributed positively to academic development and readiness for future data-analysis opportunities.

9. Bibliography

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6. Github